

GMP-Grade Enzymes for mRNA Manufacturing

Raw-material enzymes for template linearization and in vitro transcription — with matching residual-detection ELISA kits for process-impurity control.

GMP-grade product range

Workflow step	REF No.	Product	Pack size	Conc.
Template linearization	BE04103SG	BsaI, GMP Grade	20,000 U	20 U/μl
	BE04106SG	XbaI, GMP Grade	20,000 U	20 U/μl
	BE04104SG	BspQI, GMP Grade	10,000 U	10 U/μl
	BE04107SG	XhoI, GMP Grade	20,000 U	20 U/μl
	BE04105SG	NheI, GMP Grade	20,000 U	20 U/μl
In vitro transcription	BE09001SG / MG	T7 RNA Polymerase, GMP Grade	20,000 / 200,000 U	200 U/μl
	BE09011SG	Murine RNase Inhibitor, GMP Grade	40,000 U	40 U/μl
	BE09016SG	Pyrophosphatase, Inorganic (Yeast), GMP Grade	100 U	0.1 U/μl
	BE09013SG	DNase I-ST, GMP Grade (salt-tolerant, template removal)	2,000 U	2 U/μl
Residual detection	BE10002S	BsaI ELISA Kit	96 T each	—
	BE10004S	XbaI ELISA Kit		
	BE10005S	XhoI ELISA Kit		
	BE10003S	BsmBI ELISA Kit		

GMP-grade restriction enzymes and T7 RNA Polymerase are supplied with matching GMP-grade reaction buffers.

Typical GMP release specifications (example: BsaI, GMP Grade)

Test	Specification	Method
Appearance	Colorless clear liquid	Visual check
Protein purity	≥ 95%	SDS-PAGE
Non-specific endonuclease	Supercoiled plasmid conversion < 20%	20 U, supercoiled plasmid, 37 °C, 4 h
DNase activity	No dsDNA degradation observed	20 U, dsDNA fragments, 37 °C, 16 h
RNase activity	RNA degradation ≤ 10%	20 U, RNA, 37 °C, 1 h
Host-cell DNA	< 10 copies / 20 U	qPCR, residual exogenous DNA
Host-cell protein	< 50 ppm	<i>E. coli</i> HCP assay
Bacterial endotoxin	< 10 EU/mg	Bacterial endotoxins test (gel-clot)
Mycoplasma	Negative	Mycoplasma test kit
Heavy metals	< 10 ppm	Limit test for heavy metals
Microbial limit	TAMC < 5 cfu/ml; TYMC < 5 cfu/ml	Microbial enumeration

Test methods reference the Chinese Pharmacopoeia (ChP 2025, Vol. IV general chapters). Product-specific specifications are stated on each lot's Certificate of Analysis.

Quality & documentation

ISO 13485-certified QMS. Lot-specific CoA with every shipment; datasheet & SDS per product. Other documents on request.

Evaluate & scale

Samples, bulk & custom packs on request. EU stock in Warsaw, cold-chain delivery.

CPHI Milan · 6–8 Oct

Stand 10M41

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